

Note

Lab11 Exercise

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Outline

- 00_TESTBED
- 01_RTL
- 02_SYN
- 03_GATE
- 04_MEM
- 05_APR
- 06_POST
- Avoid DRC violation after nanoRoute



00_TESTBED

- Put the pattern you use in 00_TESTBED
- Include all memory.v in filelist.f

```
TESTBED.v  
../04_MEM/MEM_IMAGE.v
```

Modify this part like this if you use TA's code



01_RTL

- Use TA's RTL code
 - Prepare MEM_IMAGE.v in 04_MEM
 - Execute ./01_run_rtl
- Use your own RTL code
 - Remove the soft link named TMIP.v
 - Upload your code named TMIP.v
 - Prepare all memory.v in 04_MEM
 - Execute ./01_run_rtl



02_SYN

- Modify target_library and link_library in .synopsys_dc.setup
- Modify the synthesis cycle time in syn.tcl
- Execute 01_run_dc_shell

```

set company "iclab"
set desinger "Student"

set search_path      {
    ../01_RTL
    ../04_MEM
}

set target_library {
    fsa0m_a_generic_core_sslp62v125c.db \
    fsa0m_a_generic_core_fflp98vm40c.db \
    fsa0m_a_t33_generic_io_sslp62v125c.db \
    fsa0m_a_t33_generic_io_ttlp8v25c.db \
    MEM_IMAGE_WC.db"

set link_library      {
    * $target_library dw_foundation.sldb standard.sldb MEM_IMAGE_WC.db"
set symbol_library    {
    *.sdb "
set synthetic_library {
    dw_foundation.sldb "

set verilout_no_tri true
set hdlin_enable_presto_for_vhdl "TRUE"
set sh_enable_line_editing true
history keep 100
alias h history
  
```

The maximum speed of TA's code is 11.0 ns

Modify this part like this if you use TA's code



03_GATE

- Execute `./01_run_gate`



04_MEM

- Use TA's RTL code
 - Memory used has been generated in Memory/ftclib_200901.2.1/EXE
 - Memory name: MEM_IMAGE
 - Copy MEM_IMAGE.v / .db / .lef / .lib to 04_MEM
- Use your own RTL code
 - Prepare all_your_memory.v / .db / .lef / .lib in 04_MEM



05_APR

- Put all Memory.lef in 05_APR/LEF
- Put all Memory_WC.lib & Memory_BC.lib in 05_APR/LIB
- Follow APR steps in APR_flow_2024A.pdf
 - You can execute ./01_setenv to set OA_HOME
 - You can source apr_setting.cmd after you activate Innovus

```
05_APR]$ ./01_setenv
```

```
#!/bin/csh
```

```
setenv OA_HOME /RAID2/cad/cadence/INNOVUS/INNOVUS_20.15.000/share/oa
```

```
[INFO] Loading Pegasus 21.21 fill procedures  
innovus 1> source ./cmd/apr_setting.cmd
```

```
#####  
##### APR UMC018 setting #####  
#####  
set init_design_uniquify 1  
setDesignMode -process 180  
suppressMessage TECHLIB 1318  
suppressMessage ENCEXT-2799
```



06_POST

- Execute `./01_run_post`



Avoid DRC violation after nanoRoute

- End Iteration: more than once!
- Post Route Optimization
 - Turn on Optimize Via & Optimize Wire
- Routing Control
 - Turn on ECO Route

If you still have some DRC violation after nanoRoute with these settings, TA will suggest you should redo APR flow



Thanks

